Joint moment feedback for fixed-target acquisition

Working research note

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1 Using the joint moment circle in initial acquisition

Let P be centered with variance one and finite third absolute moment. Write

$$\rho = \mathbb{E}|X|^3, \quad \gamma = \mathbb{E}X^3, \quad \beta = \mathbb{E}[X|X|], \quad m = \mathbb{E}X_+ = \mathbb{E}X_- > 0, \quad \sigma = \sqrt{10}.$$

All moments in this note belong to this one marginal. The moment-circle inequality is already present in the manuscript's Lemma `ms:physical-defect`; the new step here is its explicit inversion and use in the initial scalar feedback. No new independent bound for the Berry–Esseen constant is claimed.

Lemma 1 (Same-law moment constraint)**.** *For $a_\pm = \mathbb{E}X_\pm^2 = (1 \pm \beta)/2$ and $b_\pm = \mathbb{E}X_\pm^3 = (\rho \pm \gamma)/2$,*

$$mb_\pm \geq a_\pm^2, \qquad m^2 \leq a_+a_-, \qquad \gamma^2 + 4m^2 \leq \rho^2. \tag{1}$$

An atom at zero is allowed.

Proof. Both signs have positive probability, since the law is centered and nondegenerate. Cauchy–Schwarz applied on each sign to $X_\pm^{1/2}, X_\pm^{3/2}$ gives $a_\pm^2 \leq mb_\pm$. If $p_\pm = P(\pm X > 0)$, a second application gives $m^2 \leq p_\pm a_\pm$. Thus

$$m^2(1/a_+ + 1/a_-) \leq p_+ + p_- \leq 1.$$

As $a_+ + a_- = 1$, this proves $m^2 \leq a_+a_-$, without assigning a zero atom a false sign. Multiplying the first two bounds gives

$$b_+b_- \geq (a_+a_-)^2/m^2 \geq m^2.$$

Since $4b_+b_- = \rho^2 - \gamma^2$, the circle follows. In particular $2m \leq 1$ and $2m \leq \sqrt{\rho^2 - \gamma^2}$. □

For completeness, use the actual gap sum J of the manuscript and its conserved defect

$$D = (3 + \sigma)\rho - \gamma - 3J.$$

Its gap bound $J \leq \rho + 2m$ remains valid before a central gap is established. Indeed, on each gap lying on one side of zero, $(b - a)^2 \leq 2\int_a^b |x|\,dx$. Summing with its nonnegative gap weight and using $2\int |x|\tau(x)\,dx = \rho$ gives the one-sided contribution. If a central gap $(-A, B)$ exists, its additional contribution is $2mAB$. Since $(X+A)(X-B) \geq 0$ on support, $0 \leq \mathbb{E}[(X+A)(X-B)] = 1 - AB$. Thus $AB \leq 1$ and the bound follows. If zero belongs to support there is no central gap. Consequently

$$D \geq \sigma\rho - \gamma - 6m \geq \sigma\rho - \gamma - 3\sqrt{\rho^2 - \gamma^2}. \tag{2}$$

The first relation is an inequality, not an alternative definition of D.

Lemma 2 (Inverting a defect ceiling)**.** *Suppose* $1 \le l \le \rho \le u$*,* $D \le d$ *with* $d \ge 0$*, and* $\sigma_l \le \sqrt{10}$*. Put* $a_l = \sigma_l - d/l$*. Define*

$$Y(a) = \begin{cases} -1, & a \le -1, \\ \dfrac{a - 3\sqrt{10 - a^2}}{10}, & -1 < a \le \sqrt{10}. \end{cases}$$

Then $\gamma/\rho \ge Y(a_l)$*. A rational lower bound* $y_l \le Y(a_l)$ *with* $y_l \ge -1$ *yields the valid cell bound*

$$\gamma \ge \begin{cases} l y_l, & y_l \ge 0, \\ u y_l, & y_l < 0. \end{cases} \tag{3}$$

Proof. Let $y = \gamma/\rho \in [-1, 1]$. Dividing (2) by ρ gives

$$f(y) := y + 3\sqrt{1 - y^2} \ge \sigma - D/\rho \ge \sigma_l - d/l = a_l.$$

The case $a_l \le -1$ imposes no new lower bound. On $[-1, 1/\sqrt{10}]$, f increases continuously from -1 to $\sqrt{10}$; its derivative in the interior is $1 - 3y/\sqrt{1 - y^2}$ and changes sign only at $1/\sqrt{10}$. Solving $f(y) = a$ on this increasing branch gives $10y^2 - 2ay + a^2 - 9 = 0$ and the stated smaller root. It has $a - y \ge 0$, including $-1 \le a \le 0$, because there $\sqrt{10 - a^2} \ge -3a$. Hence squaring introduces no spurious branch. Every y below this root has $f(y) < a$, proving the lower bound. Finally multiplying $\gamma/\rho \ge y_l$ by the positive ρ requires the endpoint l for $y_l \ge 0$ and u for $y_l < 0$. □

In exact computation the square root in Y is rounded upward, so the negative root term rounds Y downward. The branch $a_l \le -1$ keeps the whole interval. The implementation also intersects with the two previously justified bounds $\gamma \ge -g$ and $\gamma \ge \sigma_l l - 3 - d$. Thus

$$g_l = \max\{-g, \sigma_l l - 3 - d, \text{bound in (3)}\}, \tag{4}$$

$$\rho - \gamma \le \min\{2u, (1 - \sigma_l)l + 3 + d, u - g_l\}. \tag{5}$$

These inequalities retain the same actual m, ρ and γ; they do not independently optimize over moments from different laws.

1.1 Ordered raw and conserved-resource bounds

Assume the previously audited fixed-target contact/transfer identity, initial support $|X| \le 96/25$, its scalar Fourier enclosure for all $n \ge N = 42$, and $c = C_{\mathrm{E}}$. For each of the 24 closed cells

$$[r_i, r_{i+1}], \qquad r_i = 1 + 3361i/96000, \quad 0 \le i < 24,$$

start from the freshly recomputed first-pass raw ceiling T. Use rational bounds $c_- = 0.4097321837023 < c < c_+ = 0.4097321837025$ and $\phi_l = 997/2500 < \phi(0)$. The license $T < Nc_-l$ gives

$$\phi(H/\sqrt{n}) \ge \phi_l \left(1 - \frac{T}{Nc_-l}\right) =: \phi_-.$$

The previous resource ceiling U gives $D \le d = 6U/(N\phi_l)$, with outward rounding. Insert (4) and (5) into the constant charge and signed curvature of the same unsigned Fourier bound. In particular,

$$q(t) = \max\{tl - g/5,\ tl + \tfrac{1}{2}\min(\phi_- g_l, (2/5)g_l)\} > 0.$$

The constant term A, Fourier constant K_F and linear coefficient $f(t)$ otherwise keep the first pass's coherent moment and Fourier bounds. For fixed $|H|$, their upper envelope $K_F + tA + f(t)|H| - q(t)H^2$ is affine in t: the two alternatives in q have the same slope l. Therefore its maximum over $t \in [c_-, c_+]$ occurs at an endpoint before square completion.

Compute the new raw ceiling as the minimum of the old T and the outward-rounded value

$$\max_{t=c_-,c_+} \{K_F + tA + f(t)^2/(4q(t))\}.$$

The already proved conserved phase price $\eta = (c_l - T/N)/2 > 0$, formed from this *raw* ceiling, then permits

$$U_{\text{new}} = \min\left\{U, \overline{\max_{t=c_-,c_+} \{K_F + tA + f(t)^2/[4(q(t)+\eta)]\}}\right\}.$$

Bars denote upward rational rounding. Update $d_{\text{new}} = \min\{d, \overline{6U_{\text{new}}/(N\phi_l)}\}$. Every step uses the previous defect to obtain a new raw bound before obtaining a new resource and defect bound. In particular, U never replaces T in the density license. No cell is dropped or treated as a realizable law merely because its necessary constraints are consistent.

1.2 Bounded result and remaining gap

Six such updates were computed with exact fractions. All 24 cells remain covered, including shared endpoints. Their final upper bounds are

$$T \le 15.9597095614, \qquad U \le 13.2193008732, \qquad D \le 4.7353850385.$$

Twenty cells have $D < 4$ and ten have $D < 2$. The earlier coarse calculation had nineteen cells with $D < 4$, so the circle closes one of its five remaining sufficient gates. Four high-moment cells still fail that gate. The primitive Fourier/support/contact inputs remain explicit analytic dependencies; exact arithmetic does not establish them. This computation is not a completed initial acquisition, finite base, or local exclusion.